Example – 1

(a) The total pump head is given by,

$$H_p = H_{p,P} + H_{p,V} + H_{p,z}$$

$$H_{p,P} = \frac{P_d - P_s}{\gamma} \quad H_{p,V} = \frac{V_d^2 - V_s^2}{2g} \quad H_{p,z} = z_d - z_s$$

The pressure and elevation head terms can be found,

$$H_{p,P} = \frac{P_d - P_s}{\gamma} = \frac{P_d - P_s}{\rho g} = \frac{\left[250 - (-15)\right]\frac{\text{kN}}{\text{m}^3}\left|\left(1000\frac{\text{kg-m}}{\text{kN-s}^2}\right)\right.}{\left(1029.1\frac{\text{kg}}{\text{m}^3}\right)\left(9.807\frac{\text{m}}{\text{s}^2}\right)} = 26.257\ \text{m}$$

$$H_{p,z} = z_d - z_s = 0.08\ \text{m}$$

The velocity head requires the velocity at the suction and discharge of the pump,

$$V_d = \frac{4\dot{V}}{\pi D_{2.5\text{-nom}}^2} = \frac{4\left(45\frac{\text{m}^3}{\text{h}}\right)\left|\left(\frac{\text{h}}{3600\ \text{s}}\right)\right.}{\pi\left(0.052502\ \text{m}\right)^2} = 5.7739\frac{\text{m}}{\text{s}}$$

Appendix C

$$V_s = \frac{4\dot{V}}{\pi D_{2\text{-nom}}^2} = \frac{4\left(45\frac{\text{m}^3}{\text{h}}\right)\left|\left(\frac{\text{h}}{3600\ \text{s}}\right)\right.}{\pi\left(0.062713\ \text{m}\right)^2} = 4.0467\frac{\text{m}}{\text{s}}$$

Appendix C

Therefore, the velocity head is,

$$H_{p,V} = \frac{V_d^2 - V_s^2}{2g} = \frac{\left(5.7739^2 - 4.0467^2\right)\frac{\text{m}^2}{\text{s}^2}}{2\left(9.807\frac{\text{m}}{\text{s}^2}\right)} = 0.86480 \text{ m}$$

Now, the total pump head can be found,

$$H_p = H_{p,P} + H_{p,V} + H_{p,z} = (26.257 + 0.86480 + 0.08)\text{m} = \underline{27.20 \text{ m}}$$

(b) The percentage contribution of each head type to the total head is,

$$\%H_{p,P} = \frac{26.257 \text{ m}}{27.20 \text{ m}}(100\%) = \underline{96.53\%}$$

$$\%H_{p,V} = \frac{0.86480 \text{ m}}{27.20 \text{ m}}(100\%) = \underline{3.18\%}$$

$$\%H_{p,z} = \frac{0.08 \text{ m}}{27.20 \text{ m}}(100\%) = \underline{0.29\%}$$

(c) The power input to the motor driving the pump impeller is,

$$\dot{W}_p = \frac{\dot{W}}{\eta_p} = \frac{\dot{m}gH_p}{\eta_p} = \frac{\left(\rho\dot{V}\right)gH_p}{\eta_p}$$

$$\dot{W}_p = \left(\frac{1}{0.68}\right)\left(1029.1\frac{\text{kg}}{\text{m}^3}\right)\left(45\frac{\text{m}^3}{\text{h}}\right)\left(9.807\frac{\text{m}}{\text{s}^2}\right)(27.20 \text{ m})\left(\frac{\text{h}}{3600 \text{ s}}\right)\left(\frac{\text{s}^2\text{-kJ}}{1000 \text{ kg-m}}\right) = \underline{5.046 \text{ kW}}$$

(d) The annual cost required to operate the pump over the given time is,

$$c_E = (5.046 \text{ kW})\left(0.12\frac{\$}{\text{kWh}}\right)\left(8\frac{\text{h}}{\text{day}}\right)(260 \text{ day}) = \underline{\$1,259.53}$$

Example-2

The conservation of energy equation applied to a system boundary between the two free surfaces in the tanks is,

$$H_p = (z_2 - z_1) + \left(f\frac{L}{D} + \sum K\right)\frac{V^2}{2g}$$

In terms of the volumetric flow rate, this equation can be written as,

$$H_p = (z_2 - z_1) + \left(f\frac{L}{D} + \sum K \right) \frac{8\dot{V}^2}{\pi^2 g D^4}$$

The friction factor is determined from the Reynolds number,

$$\mathrm{Re} = \frac{4\rho\dot{V}}{\pi D\mu} \quad \rightarrow \quad f = \frac{0.25}{\left[\log\left(\frac{\varepsilon/D}{3.7} + \frac{5.74}{\mathrm{Re}^{0.9}} \right) \right]^2}$$

The density and viscosity of water at 60°F is (Appendix B.3)

$$\rho = 62.363 \frac{\text{lbm}}{\text{ft}^3} \qquad \mu = 0.00075334 \frac{\text{lbm}}{\text{ft-s}}$$

The inside diameter of a 4-nom pipe is (Appendix C),

$$D = 0.33550 \text{ ft}$$

Before the K values for the fittings and check valve are found, the fluid velocity must be checked to ensure that the check valve disk fully lifts.

$$V = \frac{4\dot{V}}{\pi D^2} = \frac{4(500 \text{ gpm})}{\pi(0.33550 \text{ ft})^2} \left| \left(\frac{8.02083 \text{ ft}^3}{\text{h-gpm}} \right) \left(\frac{\text{h}}{3600 \text{ s}} \right) = 12.601 \frac{\text{ft}}{\text{s}} \right.$$

The minimum velocity required to fully lift the check valve disk is (Table 4.11),

$$V_{\min} = \frac{80}{\sqrt{62.363}} = 10.130 \frac{\text{ft}}{\text{s}}$$

Since the pipe velocity is higher than the minimum velocity required, the 4-nom check valve disk will fully lift and no reducers are required to connect a smaller valve to the pipe.

The K values for the fittings and check valve can be found using the 3K method and the 2K method (for the check valve),

$$K_{elbow} = \frac{800}{\mathrm{Re}} + 0.091\left(1 + \frac{4}{D_{nom}^{0.3}} \right)$$

$$K_{entrance} = 0.5$$

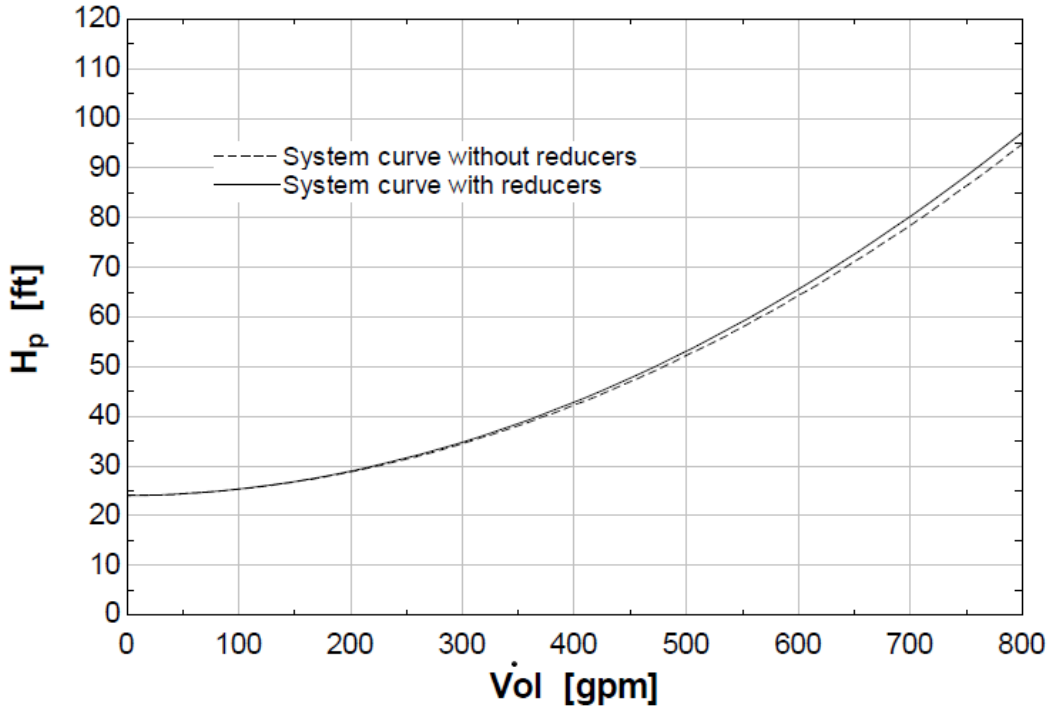
$$K_{exit} = 1$$

$$K_{check} = \frac{1000}{\mathrm{Re}} + 0.5\left(1 + \frac{1}{D[\mathrm{in}]}\right)$$

The sum of all the K values is,

$$\sum K = K_{entrance} + 3K_{elbow} + K_{check} + K_{exit}$$

Conducting a parametric study by varying the volume flow rate and calculating the system head gives the system curve,



There are two curves shown here. The dashed line is the system curve as computed with the above set of equations. At 500 gpm, the system head is about 52 ft. Consulting the pump performance maps in the Bell & Gossett Series e-1531 booklet, it appears that a model 3BD operating at 1770 rpm should work for this application. The 3BD pump has 3-inch suction and discharge ports. Therefore, reducers are required to connect this pump to the system. The K-values of the additional reducers (assumed to be rounded) are (Table 4.9),

$$K_{red} = \left(0.1 + \frac{50}{\mathrm{Re}_D}\right)\left(\frac{1}{\beta^4} - 1\right)$$

$$K_{\exp} = \left(1 + 0.8 f_d\right)\left(1 - \beta^2\right)^2$$

In these equations,

$$\beta = \frac{D_{3\,\text{nom}}}{D_{4\,\text{nom}}} = \frac{0.25567\ \text{ft}}{0.33550\ \text{ft}} = 0.76206$$

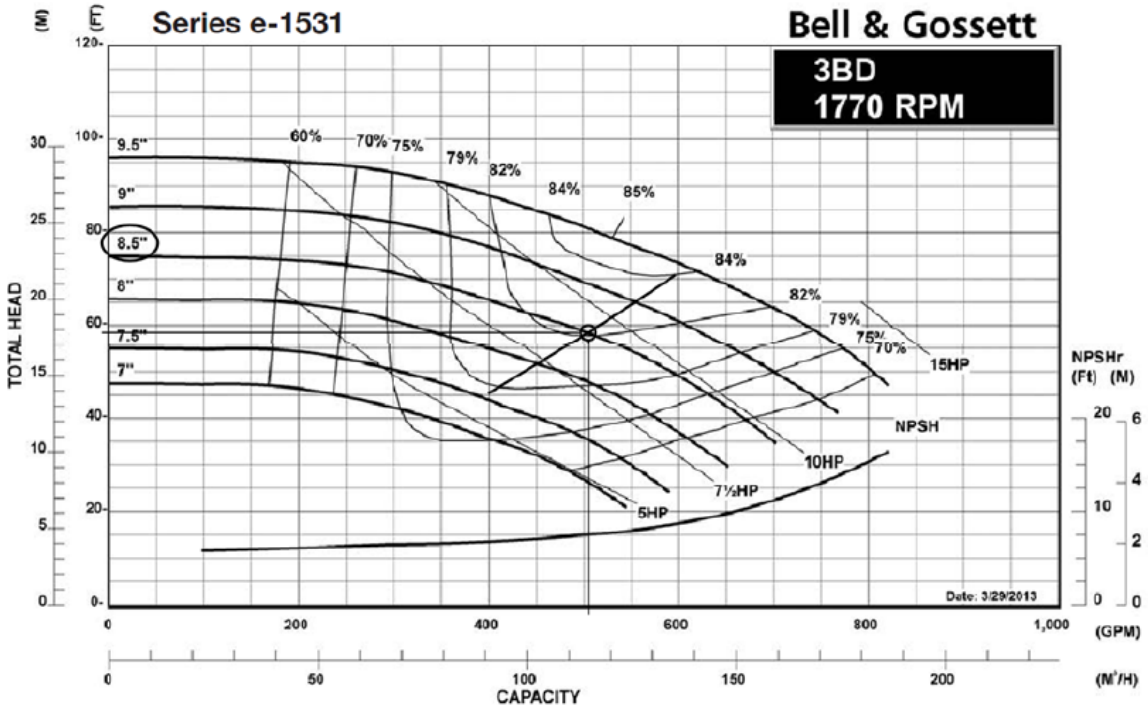
The friction factor f_d is based on the Reynold number in the smaller diameter,

$$\text{Re}_d = \frac{4\rho\dot{V}}{\pi D_{3\,\text{nom}}\mu} \quad \rightarrow \quad f = \frac{0.25}{\left[\log\left(\frac{\varepsilon / D_{3\,\text{nom}}}{3.7} + \frac{5.74}{\text{Re}_d^{0.9}}\right)\right]^2}$$

Including the K value of the reducers, the total K value is,

$$\sum K = K_{entrance} + 3K_{elbow} + K_{check} + K_{exit} + K_{red} + K_{\text{exp}}$$

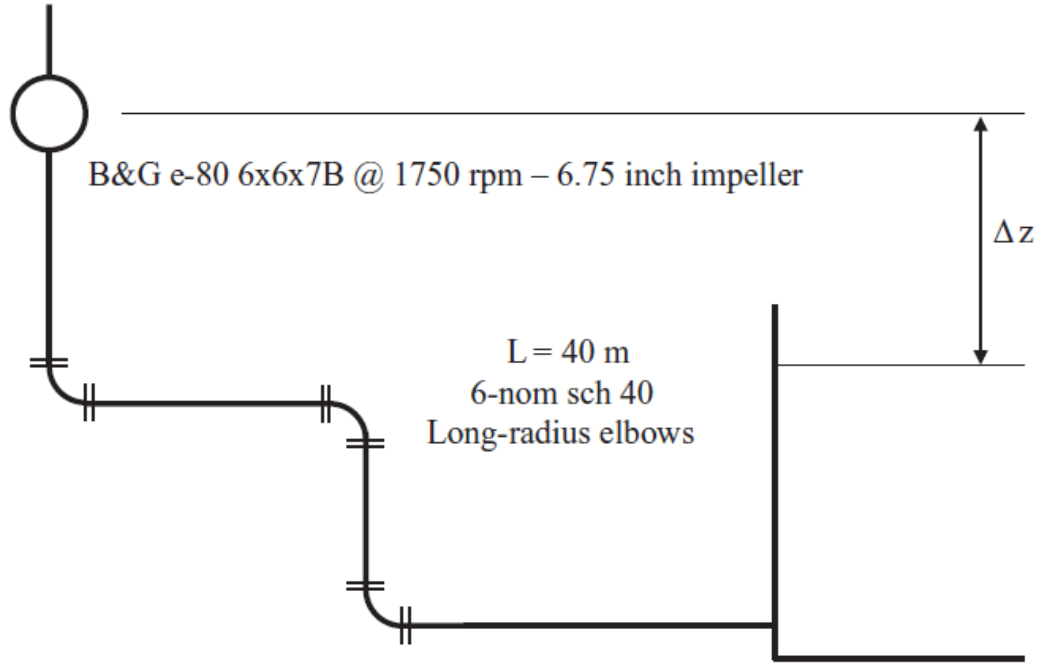
Running the parametric study again with this updated value results in the solid-line system curve shown in the previous graph. Notice that the effect of the additional resistance due to the reducers is to shift the system curve to the left. A portion of this system curve is transferred to the pump performance curves as shown below.



This indicates that the 8.5-inch impeller should be selected. This results in an operating point (indicated by the circle) of 505 gpm at 59 ft of head. The pump is operating at an efficiency of 82% and drawing about 9 hp.

Note: There may be other pumps that will work for this application. This one was selected because the operating point lies to the right of the best efficiency point (bep).

Example-3



(a) The maximum elevation above the tank surface that the pump can be placed to avoid cavitation is given by Equation (4.76),

$$z_s = z_1 + \frac{P_1 - P_v}{\gamma} - \left(f\frac{L}{D} + \sum K + 1 \right)\frac{V_s^2}{2g} - (\text{NPSHr} + \text{SF})$$

$$\Delta z = z_s - z_1 = \frac{P_1 - P_v}{\gamma} - \left(f\frac{L}{D} + \sum K + 1 \right)\frac{V_s^2}{2g} - (\text{NPSHr} + \text{SF})$$

The properties of water at 20°C are (Appendix B.1)

$$\rho = 998.16 \frac{\text{kg}}{\text{m}^3} \quad \left[\gamma = \rho g = 9789.0 \frac{\text{N}}{\text{m}^3} \right] \quad \mu = 0.0010016 \frac{\text{kg}}{\text{m-s}} \quad P_v = 2.3393 \text{ kPa}$$

The pipe data is determined from Appendix C and Table 4.3,

$$D = 6.065 \text{ in} = 0.15405 \text{ m} \qquad \varepsilon = 0.000046 \text{ m}$$

The NPSHr can be read from the pump's performance curve. At 160 m^3/h,

$$\text{NPSHr} = 3.4 \text{ m}$$

The saftey factor, SF = 1 m.

The friction factor is found by,

$$V = \frac{4\dot{V}}{\pi D^2} = \frac{4\left(160 \frac{\text{m}^3}{\text{h}}\right)\left(\frac{\text{h}}{3600 \text{ s}}\right)}{\pi (0.15405 \text{ m})^2} = 2.3845 \frac{\text{m}}{\text{s}}$$

$$\text{Re} = \frac{\rho V D}{\mu} = \frac{\left(998.16 \frac{\text{kg}}{\text{m}^3}\right)\left(2.3845 \frac{\text{m}}{\text{s}}\right)(0.15405 \text{ m})}{\left(0.0010016 \frac{\text{kg}}{\text{m-s}}\right)} = 366,067$$

$$f = \frac{0.25}{\left[\log\left(\frac{\varepsilon / D}{3.7}\right) + \frac{5.74}{\text{Re}^{0.9}}\right]^2} = \frac{0.25}{\left[\log\left(\frac{0.000046 \text{ m} / 0.15405 \text{ m}}{3.7}\right) + \frac{5.74}{366,067^{0.9}}\right]^2} = 0.016755$$

The minor losses are determined using the 2K method. (Note: The 3K method does not have constants listed for a flanged long-radius elbow).

$$K_{elbow} = \frac{800}{\text{Re}} + (0.4)\left(1 + \frac{1}{D[\text{in}]}\right) = \frac{800}{336,067} + (0.4)\left(1 + \frac{1}{6.065[\text{in}]}\right) = 0.23516$$

$$K_{entrance} = \frac{160}{\text{Re}} + 0.5 = \frac{160}{336,067} + 0.5 = 0.50044$$

$$\therefore \quad \sum K = 3(0.23516) + 0.50044 = 1.20592$$

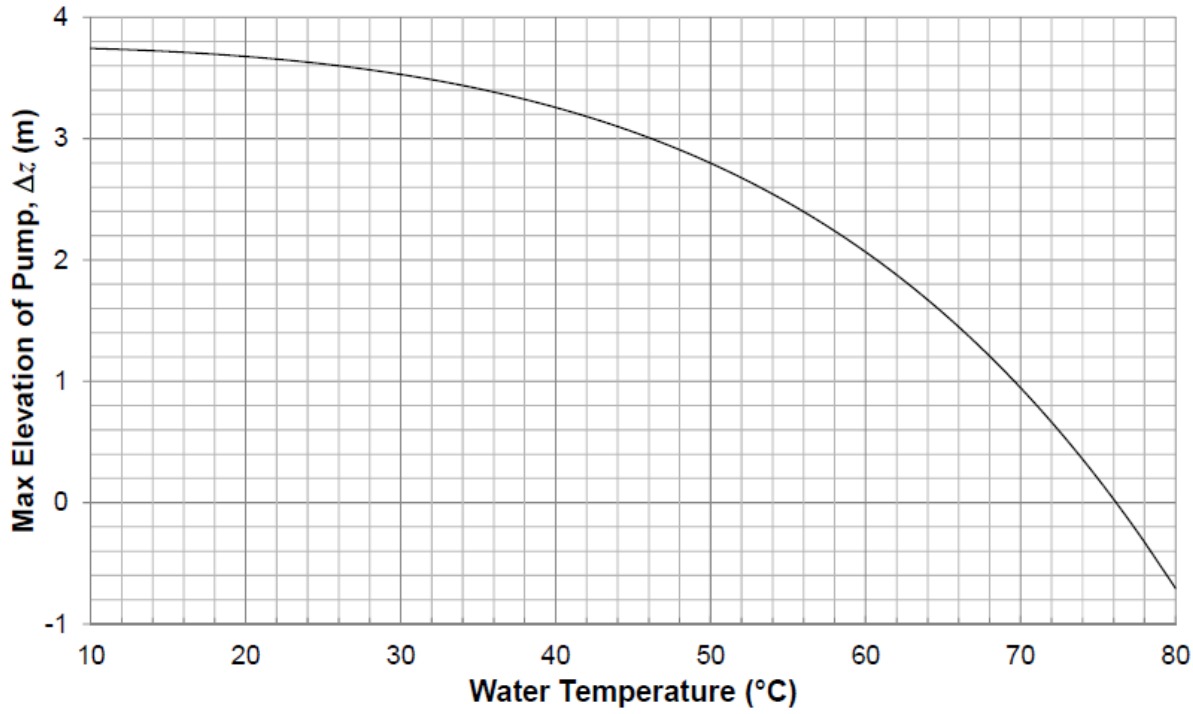
Therefore, the maximum elevation that the pump can be installed to avoid cavitation is,

$$\Delta z = \frac{\left[1\ \text{bar}\left(\frac{100\ \text{kPa}}{\text{bar}}\right) - 2.3393\ \text{kPa}\right]\left(\frac{1000\ \text{N}}{\text{m}^2\text{-kPa}}\right)}{9789.0\frac{\text{N}}{\text{m}^3}}$$

$$-\left[0.016755\left(\frac{40\ \text{m}}{0.15405\ \text{m}}\right)+1.20592+1\right]\frac{\left(2.3845\frac{\text{m}}{\text{s}}\right)^2}{2\left(9.807\frac{\text{m}}{\text{s}^2}\right)}-(3.4+1)\text{m}$$

$$\Delta z = \underline{3.68\ \text{m}}$$

(b) A parametric study, varying T and calculating Δz can be done using computer software. Using MS Excel, the result is shown below.



This plot shows that the temperature of the water has a significant impact on the maximum elevation of the pump for cavitation prevention. Notice that at temperatures higher than 76°C, the pump must be located *below* the surface elevation of the water.

Example-4

The dimensionless groups used to compute the capacity, head, and power draw under the new condition are,

$$\frac{\dot{V}_n}{\omega_n D_n^3} = \frac{\dot{V}_o}{\omega_o D_o^3} \qquad \frac{gH_{p,n}}{\omega_n^2 D_n^2} = \frac{gH_{p,o}}{\omega_o^2 D_o^2} \qquad \frac{\dot{W}_n}{\rho_n \omega_n^3 D_n^5} = \frac{\dot{W}_o}{\rho_o \omega_o^3 D_o^5} \qquad \eta_n = \eta_o$$

Therefore,

$$\dot{V}_n = \dot{V}_o \left(\frac{\omega_n}{\omega_o}\right)\left(\frac{D_n}{D_o}\right)^3 = \left(50\frac{\text{L}}{\text{s}}\right)\left(\frac{1150\text{ rpm}}{3550\text{ rpm}}\right)\left(\frac{9\text{ in}}{7\text{ in}}\right)^3 = \underline{34.4\frac{\text{L}}{\text{s}}}$$

$$H_{p,n} = H_{p,o}\left(\frac{\omega_n}{\omega_o}\right)^2\left(\frac{D_n}{D_o}\right)^2 = (18\text{ m})\left(\frac{1150\text{ rpm}}{3550\text{ rpm}}\right)^2\left(\frac{9\text{ in}}{7\text{ in}}\right)^2 = \underline{3.12\text{ m}}$$

$$\dot{W}_n = \dot{W}_o\left(\frac{\omega_n}{\omega_o}\right)^3\left(\frac{D_n}{D_o}\right)^5 = (12\text{ kW})\left(\frac{1150\text{ rpm}}{3550\text{ rpm}}\right)^3\left(\frac{9\text{ in}}{7\text{ in}}\right)^5 = \underline{1.43\text{ kW}}$$

Example-5

(a) There are two points on the system curve that are known; (0 gpm, 0 ft) and (400 gpm, 75 ft). Since the system curve is a parabola symmetric about the y-axis, the equation for the system curve can be written as,

$$H_p = C\dot{V}^2$$

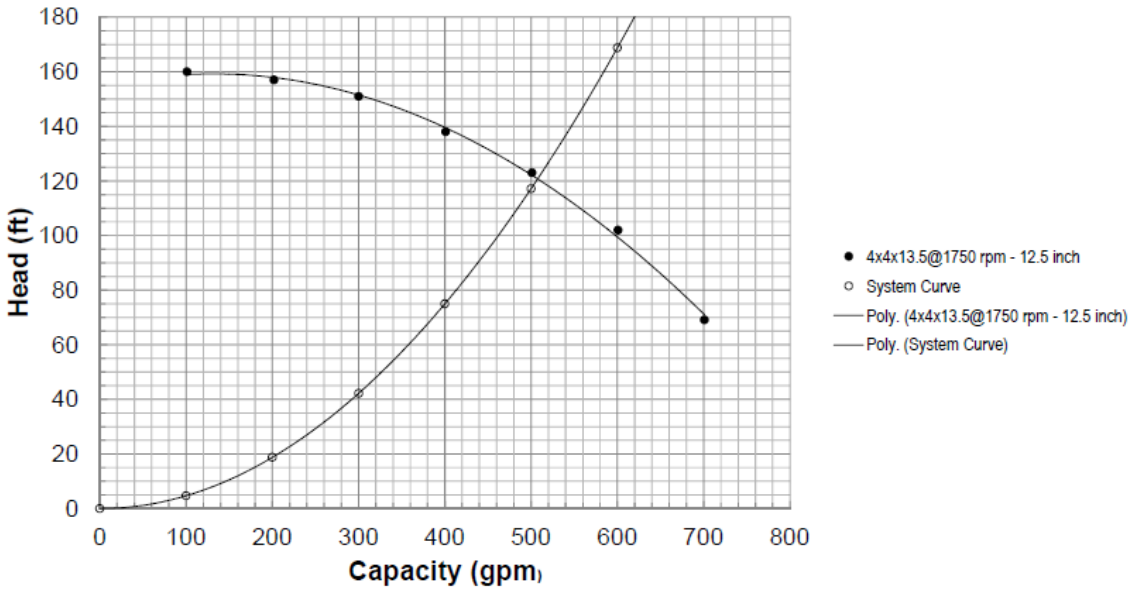
The value of C is constant along the system curve. Therefore the system curve is be represented by,

$$C = \frac{H_p}{\dot{V}^2} = \frac{75 \text{ ft}}{(400 \text{ gpm})^2} = 0.00046875 \frac{\text{ft}}{\text{gpm}^2}$$

$$\therefore \quad H_p = 0.00046875 \dot{V}^2$$

Data points from the pump performance curve can be read and plotted. The resulting data set and plot showing the pump curve and system curve is shown below.

original conditions	
V_dot (gpm)	H_p (ft)
101	160
202	157
300	151
401	138
501	123
601	102
701	69.1



The operating point can be read directly from this plot,

$$\dot{V}_o = \underline{507 \text{ gpm}} \qquad H_{p,o} = \underline{121 \text{ ft}}$$

(b) Using the affinity laws to find the corresponding pump performance characteristics at the new operating conditions,

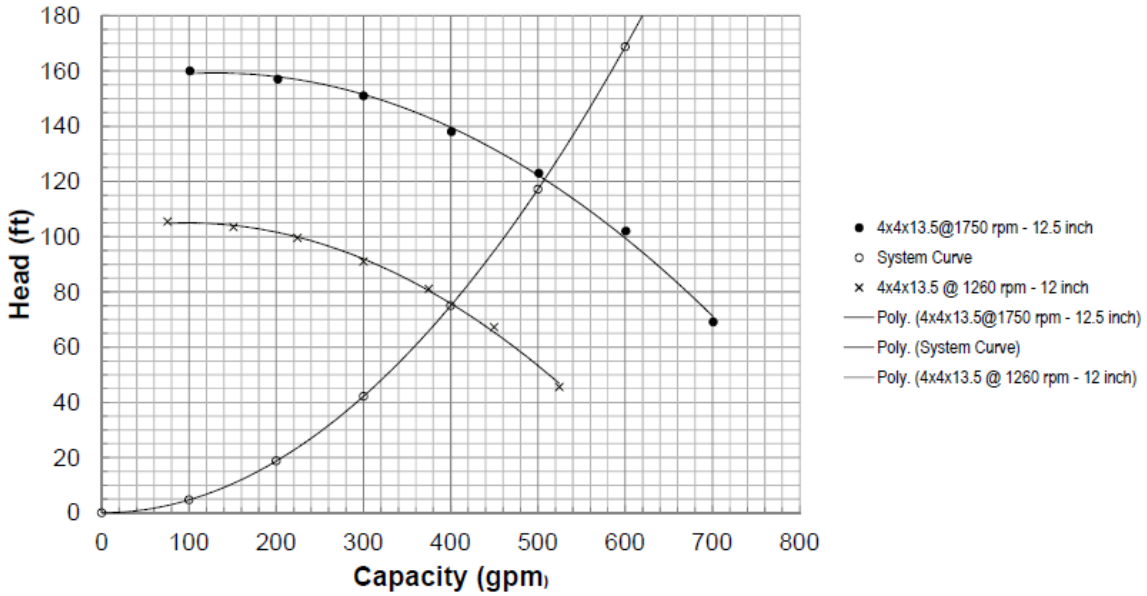
$$\dot{V}_n = \dot{V}_o\left(\frac{\omega_n}{\omega_o}\right)\left(\frac{D_n}{D_o}\right)^3 = (507\text{ gpm})\left(\frac{1480\text{ rpm}}{1750\text{ rpm}}\right)\left(\frac{12\text{ in}}{12.5\text{ in}}\right)^3 = \underline{379\text{ gpm}}$$

$$H_{p,n} = H_{p,o}\left(\frac{\omega_n}{\omega_o}\right)^2\left(\frac{D_n}{D_o}\right)^2 = (121\text{ ft})\left(\frac{1480\text{ rpm}}{1750\text{ rpm}}\right)^2\left(\frac{12\text{ in}}{12.5\text{ in}}\right)^2 = \underline{80\text{ ft}}$$

(c) Applying the affinity laws to the data points read in part (a) results in the following set of new conditions,

original conditions		new conditions	
V_dot (gpm)	H_p (ft)	V_dot (gpm)	H_p (ft)
101	160	75.57	105.47
202	157	151.14	103.49
300	151	224.47	99.53
401	138	300.04	90.96
501	123	374.87	81.08
601	102	449.69	67.23
701	69.1	524.51	45.55

These points can be added to the plot previously generated to indicate the new operating point as shown below.



This plot shows that the new operating point of the pump/pipe system is,

$$\dot{V}_o = \underline{400\text{ gpm}} \qquad H_{p,o} = \underline{75\text{ ft}}$$

(d) The two points calculated in parts (b) and (c) do not match because the point calculated in part (b) is on the pump performance curve whereas the new operating point calculated in part (c) is the new operating point. The affinity laws do not scale pump/pipe system operating points. They are used to scale pump performance curves.